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**Indian Institute of Information Technology Surat**  
भारतीय सूचना प्रौद्योगिकी संस्था सुरत  
(An Institute of National Importance under Act of Parliament)

*Presentation on*  
**Multi-scale Adaptive Graph Network for  
Spatio-Temporal Forecasting**

***Group members***

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# Introduction:

- Urban traffic is complex and constantly changing, making vehicle speed prediction difficult.
- We model the city road network as a graph, where nodes represent speed sensors on roads. Our approach combines **Dynamic Adjacency Matrix**, **GCN** for spatial features, **TCN** for temporal patterns, and **Cross-Attention** to improve feature interaction.

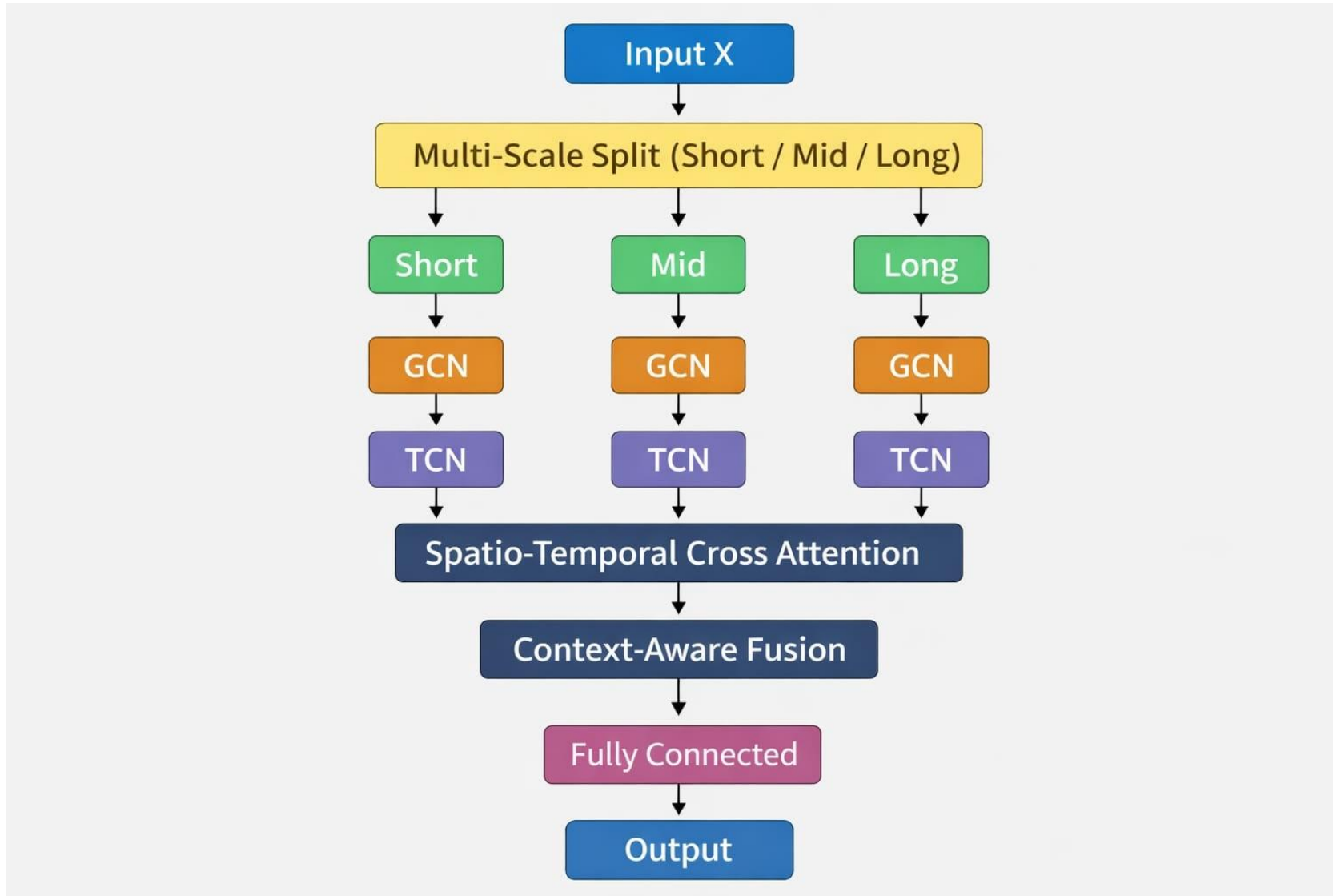
# Literature Survey:

- **Graph WaveNet (*TCN + Graph Convolution*)**
  - Combines Temporal Convolutional Networks (TCN) with adaptive Graph Convolution
  - Learns hidden spatial dependencies automatically
  - Handles long-range temporal patterns efficiently
- **STGCN (*Spatio-Temporal Graph Convolutional Network*) (*Graph Conv + Temporal Conv*)**
  - Integrates graph convolution for spatial features
  - Uses temporal convolution for sequential data
  - Efficient and fully convolutional (no RNNs)
- **STAHGCN (*GCN + TCN + Attention-Based Learning*)**
  - Enhances GCN + TCN with attention mechanism
  - Focuses on important nodes and time steps
  - Improves performance on complex dynamic graphs
- **ASTGCN (*Multi-Scale: Short, Mid, Long Term*)**
  - Captures temporal patterns at different time scales
  - Uses attention mechanisms for spatial-temporal features
  - Suitable for traffic forecasting and time-series tasks

# Problem Statement:

- Accurate prediction of average vehicle speed in urban road networks is challenging due to dynamic traffic conditions, complex spatial dependencies between roads, and temporal variations over time.
- Existing methods often fail to adapt to changing traffic patterns and do not effectively capture both spatial and temporal relationships.
- Therefore, there is a need for a robust model that can dynamically learn road connectivity and accurately predict vehicle speed at each node in a city-wide traffic graph.

# Proposed Model:



# Requirements:

- **Programming Language: Python**
  - Easy to learn, readable, and widely used
  - Strong ecosystem for AI/ML development
  - Supports rapid prototyping
- **Framework: TensorFlow**
  - Open-source library for machine learning and deep learning
  - Supports neural networks and large-scale models
  - Optimized for GPU/TPU acceleration
- **Data Processing: NumPy & Pandas**
  - **NumPy:** Fast numerical computations, array operations, linear algebra
  - **Pandas:** Data manipulation, cleaning, and analysis using DataFrames
- **Visualization: Matplotlib & Seaborn**
  - **Matplotlib:** Basic plots (line, bar, scatter) with high customization
  - **Seaborn:** Advanced statistical visuals with better design and simplicity

# Applications:

- **Route Planning & Navigation** – Improve travel time estimation in apps
- **Smart Traffic Management** – Optimize signal timing and reduce congestion
- **Smart City Systems** – Support real-time urban mobility decisions
- **Logistics & Fleet Management** – Enhance delivery efficiency and planning
- **Accident & Congestion Prediction** – Enable proactive traffic control
- **Public Transport Optimization** – Improve scheduling and reliability

# Results

| MODEL     | MSE      | RMSE   | MAE    |
|-----------|----------|--------|--------|
| LSTM      | 43.125   | 6.567  | 2.8252 |
| GRU       | 43.053   | 6.561  | 2.8182 |
| STGCN     | 43.17    | 6.57   | 2.8242 |
| HA        | 3106.66  | 55.737 | 50.82  |
| ARIMA     | 3045.772 | 55.189 | 49.987 |
| MAGNet-ST | 37.544   | 6.127  | 2.578  |

# Future Scope:

- **Real-time data integration** – Use live traffic, GPS, and sensor feeds for better accuracy
- **External factors** – Incorporate weather, events, and accidents into the model
- **Adaptive learning** – Continuously update the model with new incoming data

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